

Enterprise-Grade Recommendation System

Module Guides — index and concept map

What these documents are

One guide per module. Each explains what the module does in plain language, lists every concept it uses, walks through how it works step by step, shows the key code, flags the things worth remembering, and ends with the questions most likely to be asked about it.

The 14 modules

#	Module	File	Lines	What it does
01	Synthetic Data Generator	generate_synthetic_data.py	751	Creates the entire dataset: 6,000 customers, 2,500 products, 138,345 interactions.
02	Data Loader	data_loader.py	70	Reads the three CSV files. The smallest module, with one line that prevents a serious bug.
03	Preprocessing & Feature Engineering	preprocessing.py	279	Turns the raw interaction log into the tables and matrices the models need.
04	Content-Based Recommendation (NLP)	content_based_nlp.py	313	Recommends products by their text, which is how new products get recommended at all.
05	Collaborative Filtering Improvements	collaborative_filtering.py	282	Fixes the specific ways textbook collaborative filtering fails on real data.
06	Baseline Recommenders	baseline_recommenders.py	283	The four mandated baselines. These set the bar every other model must clear.
07	Neural Collaborative Filtering (PyTorch)	ncf_recommender.py	609	The deep learning model - and the single most important lesson in the project.
08	Hybrid Recommender	hybrid_recommender.py	411	Combines four signals into one ranked list. This is the best-performing model.
09	Explainability Layer	explainability.py	397	Answers 'why was this recommended?' - worth 10 marks in the rubric.
10	Evaluation Metrics	evaluation_metrics.py	198	The ranking metrics, written from scratch rather than imported.
11	EDA & Data Validation	eda_validation.py	337	Proves the generated data actually behaves like real shopping data.
12	Benchmark Harness	recommendation_evaluation.py	462	Compares every model fairly. This is where the two biggest bugs were found.
13	FastAPI Service (Backend)	app_fastapi.py	465	Serves recommendations over HTTP. Transport only - no scoring logic lives here.
14	Streamlit Dashboard (Frontend)	streamlit_recommendation_dashboard.py	551	The visual console - built for whoever has to explain a recommendation.

Headline results

Model	NDCG@10	Coverage	Notes
Hybrid	0.1095	20.2%	best on every ranking metric
Item-based CF	0.1044	22.2%	strongest single model
NCF	0.0663	1.3%	implicit variant with negative sampling
Popularity	0.0645	1.3%	the baseline to beat
SVD	0.0462	9.6%	generalises across sparse data
Content	0.0116	75.4%	weakest accuracy, best reach

The hybrid beats the popularity baseline by **+69.9%** on NDCG@10, measured on a time-based split with leakage assertions and every model refitted on training data only.

Dataset

Property	Value
Customers	6,000
Products	2,500
Interactions	138,345
Sparsity	99.08%
Popularity concentration	top 10% of products hold 62.9% of demand
Long tail	Gini 0.716
Cold-start customers	480 (8.0%)
Cold-start products	255 (10.2%)
Mean rating	3.91, J-shaped